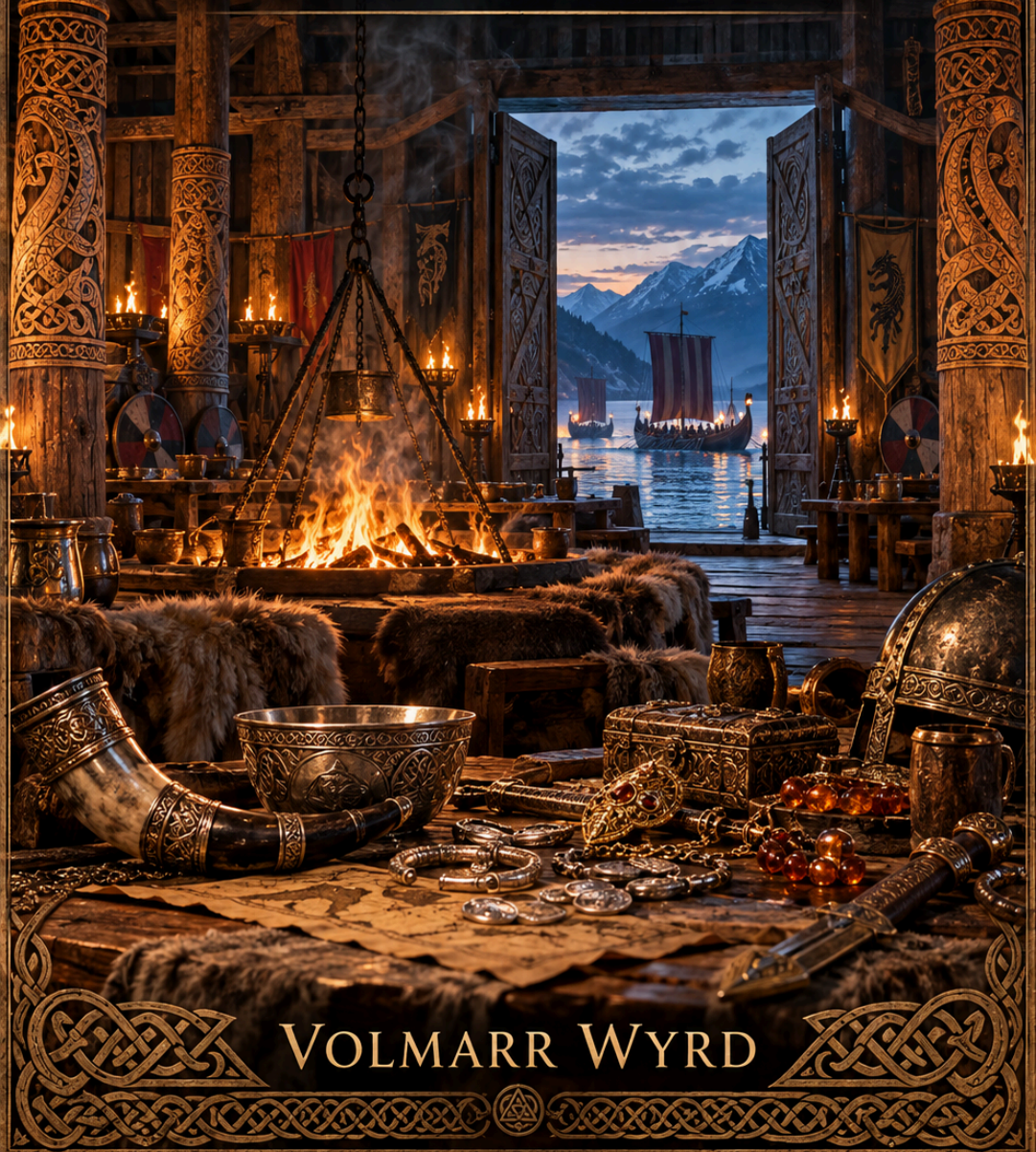


Smoke, Silver & Sea

A LIVING HISTORY OF THE VIKING AGE



VOLMARR WYRD

SMOKE, SILVER & SEA

A Living History of the Viking Age

793–1066 CE

Volmarr Wyrð

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A Living History of the Viking Age



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Volmarr Wyrð

First published edition, 2026

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Copyright

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Smoke, Silver & Sea: A Living History of the Viking Age

First published edition, 2026.

This is a research-based work of living history: twenty-five chapters of Viking Age life built from academic sources, including excavation reports, peer-reviewed studies, museum publications, inscriptions, contemporary testimony, and identified translations of medieval texts. Its governing principle is simple: preserve every good story, but never preserve a certainty the evidence cannot bear.

Composite characters are clearly presented as representative figures rather than historical persons. Archaeology, contemporary testimony, later medieval literature, reconstruction, and speculation are distinguished wherever the surviving evidence permits. Every block quotation is attributed to a named source or edition in the notes, and disputes over chronology, translation, or interpretation are stated rather than hidden behind false certainty.

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Sample excerpt: front matter, method, introduction, and Chapter 1.

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Sample edition omits the Preface and Acknowledgments.

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How We Know --- A Note on Method and Evidence

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This book begins with dirt. Before the kings, before the sagas, before the famous names, there are postholes, bones, slag, rivets, graves, harbour mud, inscriptions, and things people dropped and never came back for. Archaeology comes first; the stories come after. The sources throughout are university publications, peer-reviewed papers, excavation reports, national museum collections, and established scholarship. When the book gives a date, dimension, count, or find, the chapter notes show where it came from.

A few principles governed the work throughout:

The spade before the saga. The sagas and poems written down in thirteenth-century Iceland are magnificent literature, and this book quotes them — but always as literature, in real identified translations, never as eyewitness reporting on the ninth century. When the archaeology and the stories disagree, the archaeology wins. When they agree, the stories get to sing.

No invented things. No invented quotations, no invented statistics, no invented citations. Every blockquote in these pages is a genuine passage from a named source in a named translation. Where a figure is a reconstruction or a range, it is labeled as one.

Disputes are shown, not hidden. The Viking Age is an argumentative field, and this book does not pretend otherwise. The woman in Birka grave Bj 581, the origins of the Rus, the causes of the raiding, the temple at Uppsala — on each, the honest state of the debate is laid out with the evidence for every side, and the reader is trusted to weigh it.

“Viking” is used with care. *Víkingr* in Old Norse meant a raider, a pirate — an occupation, not a people. The people of this book are Norse, Danes, Swedes, Norwegians, Icelanders, Rus, Gall-Gaedhil: farmers who raided, traders who farmed, kings who did both. The book says “Viking Age” for the period, because that is the convention, and it says “Norse” for the culture, because that is closer to the truth.

Uncertainty is stated precisely. “We do not know” appears in these pages as a finding, not a failure. The difference between what is at-

tested, what is reconstructed, and what is guessed is the difference between history and fantasy, and this book keeps the three apart.

People are described inside their own world. Viking Age societies are not ranked on a ladder from “primitive” to “advanced,” and present-day values are not used as a scoreboard for people living under different legal, religious, economic, and ecological conditions. Where the evidence shows coercion, violence, enslavement, hierarchy, suffering, or conflict, those conditions are described concretely and in their historical context. The aim is neither condemnation nor exoneration, but understanding.

No book can contain the whole Viking Age. What follows is a living history of the land, towns, roads, crafts, silver, swords, gods, and law — told through what the earth has given back and what the age, cautiously and imperfectly, said about itself. The spade gets the first word. When the stories can carry their weight, they get to sing.

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Introduction --- The Age Between Two Worlds

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Picture the monks at their prayers.

It is the 8th of June, 793, on the island of Lindisfarne off the Northumbrian coast — a holy place where the North Sea wind carries the salt tang of the world's edge. The brothers are bent over their manuscripts, quills scratching vellum, when the sound reaches them: not the familiar crash of waves, but something else. Something unfamiliar. Raiders from the north are on the beach.

The Anglo-Saxon Chronicle's entry for that year remembers it as the coming of "the ravaging of wretched heathen men." And the scholar Alcuin, writing to a Northumbrian king, reached for words the age had not needed before: never, he said, had such terror been seen in Britain.

Two hundred and seventy-three years later, the conventional span closes on a battlefield in the north of England. On the 25th of September, 1066, Harald Hardrada of Norway, the last great Viking king-adventurer, fell at Stamford Bridge in Yorkshire. Nineteen days after that, another descendant of Vikings, William of Normandy, won England at Hastings. Between those two bookends lies the Viking Age.

It is a conventional span, and like all conventions it can be argued with. The raiding did not start at Lindisfarne — there were earlier attacks, and the trading towns of the north were already stirring — and Norse life did not stop in 1066; it changed its clothes, took baptism in larger numbers, and went on farming the same fjords. But 793 to 1066 remains a useful frame: the centuries when people speaking Old Norse expanded through trade, settlement, raiding, service, and migration across a world stretching from Newfoundland to Baghdad, from Dublin to Kyiv.

What "Viking" means

The word itself is the first trap — so let's spring it now. In Old Norse, *vikingr* meant a raider, a sea-robber. A job description, not an ethnicity. To "go viking" (*fara í viking*) was to go on expedition, the way you might

go on campaign or go to sea. Most of the people in this book never went viking at all.

Meet the people who actually filled the age. The *bœndr*, farmers, who measured wealth in livestock and counted the year by the lambing. The *smiðir*, smiths, standing in the smoke of their forges turning bog ore into blades. The *kaupmenn*, merchants, who knew the sandbars of a dozen harbors by touch. The *húsfreyjur*, household-mistresses, with iron keys chiming at their waists — the keys to the farm’s wealth, and the power that went with them. The raiders were the sharp edge of a whole society, and this book is about the whole society. The edge makes no sense without the blade behind it.

That society was Norse — Danish, Norwegian, Swedish, and the Norse settlers of Iceland, the Faroes, Orkney, Ireland, England, Normandy, and the Rus river-routes. They shared a language family, a material culture, a set of gods, and a legal imagination. They did not share a nation, a king, or a plan. Here is the thing most retellings get wrong: **there was never a Viking empire and never a Viking master strategy.** There were farms, towns, ships, ambitions, and opportunities — and three centuries of consequences.

How we know

We know the Viking Age the way we know anything buried: in fragments, and more completely than you might expect. The evidence comes in several kinds, and each kind lies in its own way — which is why this book always tells you which kind is talking.

The earth. Stand in a waterlogged harbor layer at Hedeby or Birka or Kaupang, in York or Dublin, and hold what the mud refused to rot: wood, leather, cloth, food. Graves give back swords, brooches, ships — and skeletons, whose bones and teeth now yield diet, disease, and migration histories through isotope analysis and ancient DNA. Hoards give back silver by the kilogram, and with it the trade routes mapped in metal. The archaeology is the backbone of this book. Where it speaks clearly, it overrules every story.

The stones. Thousands of runestones stand across Scandinavia, most raised in the eleventh century by Christians who still thought in Norse. They name the dead. They name the voyages east to “Greece” — Byzantium. They name the men who “fell in the east,” and the widows who raised the stones. Terse, formulaic, contemporary: the closest thing the age left to its own voice in its own script.

The neighbors’ accounts. The Franks, the Anglo-Saxons, and the Irish wrote about the raiders at length — usually while being raided, which colors the testimony. The Arabic travelers wrote as curious out-

siders, which colors it differently: Ibn Fadlan on the Volga, al-Tartushi at Hedeby. The Byzantines hired Norse guardsmen and wrote down their pay. Every one of these witnesses had an agenda. This book names them.

The Norse words themselves. Skaldic verse, composed for kings in the tenth and eleventh centuries and preserved in later manuscripts, is the oldest Norse literature we have. The sagas and the Edda, written down in thirteenth-century Iceland — two centuries after the age ended — are the richest and the most source-critical: vivid, detailed, and shaped by the concerns of Christian Icelanders looking back at a pagan past. This book drinks from them but labels the cup.

The land. The map remembers what the chronicles forgot. Place-names ending in *-by*, *-thorpe*, *-thwaite* scattered across England. *Nor-mandy* itself. The Danelaw's boundaries. The farm mounds of Norway, still farmed today.

The shape of this book

The book moves the way the Norse world worked: from the inside out. It begins with the land and the farmstead — because nine out of ten Norse people lived and died on farms, and everything else, the towns, the ships, the silver, the swords, was built on the farm's surplus. Then it walks through the trading towns west and east. It follows the iron from bog to blade, the timber from forest to ship, the silver from the Caliphate's mints to Scandinavian hoards. Then it turns to war: the raiding and the great armies, treated within the same political and economic systems as trade, settlement, tribute, and kingship. And it ends where Norse life ended and began again — the holy places, the rites of passage, the law, and the turning year.

What this book will not do

It will not give you horned helmets. No Viking Age helmet with horns has ever been found. The one nearly complete helmet we have, from Gjermundbu in Norway, is a plain iron cap with a spectacle guard — workmanlike, unadorned, real.

It will not give you a single “Viking religion” or “Viking law,” as if the whole Norse world believed and judged identically from Lofoten to Kyiv. It will not separate raiding and trading into tidy moral categories. The same ships, ports, households, and political networks could support commerce, warfare, tribute, migration, and the traffic in enslaved peo-

ple. The slave trade was structurally embedded in parts of the Viking Age economy, and it is treated here as part of that economy rather than as an exception to it.

What it will give you is the age as the evidence allows: the smell of the longhouse fire, the weight of the hacksilver, the geometry of the clinker nail, the words of the oath, the silence of the bog where the weapons were laid down.

Three centuries. Twenty-five chapters. One world — as close as scholarship can bring you to standing in it. Come and stand in it.

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PART I

Land and Farmstead

...

Before there were Vikings, there were farmers. Everything in this book — the ships, the silver, the swords, the songs — was paid for by barley, butter, and fish, grown and caught by people who measured wealth in livestock and counted the year by the lambing. Start here, in the longhouse, by the central fire, and the rest of the age will make sense.

CHAPTER 1

Land & Sea

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Dawn on the Inner Lead

An hour before sunrise, the fjord holds its breath. You are in the stern of the boat.

The water lies dark and oiled beneath you, carrying the fellside's reflection in one unbroken sheet. The first grey light comes down the water from the open sea in the west, feeling its way along the walls. Your boat is small — a karvi, or a fishing boat — low in the water, her oars shipped because there is no wind yet and the helmsman knows this passage the way you know the rooms of your own house: by touch. She threads between skerries, grey humps of rock with gulls sleeping on them, along a channel that lies calm while the open sea beyond the outer islands is white with surf.

This is the inner lead: the sheltered road between the skerry chain and the mainland, running north and south along the whole length of Norway's coast. On a morning like this, with the fjord glassy and the tide

slack, it is the easiest traveling in the medieval world. The steersman needs no chart. He needs the marks — the cleft in that headland that means the deep water bends left, the reef that shows only at low water, the farm at the head of the next bay where a man can beach for the night and be fed. He learned them from his father, who learned them from his.

Before the sun is up the ship is past the narrows and into the wider fjord, and the land opens around her. The walls rise a thousand metres and more on either side, streaked with waterfalls that are silver threads in the half-light. Near the water the slopes are forested — pine and spruce, dark green even in winter, the resin smell sharp on the cold air. Higher, the forest thins and stops, and above the treeline the fell lies bare: blue-grey rock, old snow in the gullies. And at the head of the fjord, invisible from here, a flat tongue of green marks where a river has laid down its silt — a farm, perhaps two, a church or a burial mound, a strip of barley field no wider than a bowshot. That is where the people live. Not on the grand sweep of the fjord, which belongs to water and rock, but in the thin, negotiated places where the land relents.

Everything the Viking Age became — the ships, the outward looking, the farms that faced the water rather than the road, the wealth that flowed from dried fish and sea trade — begins in this relationship between a people and a geography that gave them almost no choice but the sea. This chapter is about that geography: the shape of the land, the climate that scholars have been arguing over for a century, the two different worlds of coast and inland, and the ocean road that turned a scattering of farmsteads into a connected world.

The Shape of the Land

Fjords, Fjells, Forests

Picture the ice. Not the ice you know — not winter frost, not skating ponds — but ice kilometres thick, pressing the very bedrock down into the earth's mantle like a thumb on dough. Scandinavia is, first and last, a glaciated land, and the last ice sheet only finished releasing its grip some ten thousand years ago. A blink, in geological time. It left behind a landscape still marked by its passage: scoured bedrock, erratic boulders dropped where the meltwater abandoned them, and above all the fjords.

A fjord is a drowned glacial valley. During the ice ages, glaciers ground their way down river valleys to the sea, gouging them into deep, steep-sided troughs — far deeper than any river could cut, and characteristically U-shaped in profile, with flat floors and abrupt walls. When

the ice melted and the sea rose, the ocean flooded back up these troughs. Consider the Sognefjord in western Norway: 205 kilometres of inland reach, plunging to 1,308 metres below the surface — the longest and deepest fjord in the country, the second-longest in the world.¹ In places its floor lies hundreds of metres beneath sea level while farms cling to ledges a thousand metres above it. And the land itself, having been pressed down by the ice, is still rebounding — rising millimetre by millimetre out of the sea. The Vikings lived in a landscape that was, in a literal sense, still surfacing.

Above the fjords rise the fjells — the high mountain plateaus and barren highlands that make up the spine of Norway and much of inland Sweden. The word *fjell* covers everything from forested uplands to treeless rock and permanent snow. Between the fjells and the sea lie the forests: pine, spruce, and birch, in the Viking Age far more extensive than today, reaching down in places almost to the water's edge. The great inland forests of Sweden and eastern Norway were a world of their own — dark, trackless, a frontier that swallowed travellers and sheltered outlaws. Imagine the smell of that forest: resin and rot and cold leaf-mould, the silence broken only by an axe-stroke or a woodpecker's hammer.

And then there is the farmland: the exception, not the rule. Of all the land in Norway, only about three percent is suited to cultivation or arable farming; the rest is mountain, forest, heath, and bog.² In the fjord districts of western Norway the habitable areas at the bottom of the steep mountainsides are, in the FAO's dry phrasing, "very limited" — a ribbon of soil between cliff and water, sometimes only a few dozen metres wide, where a farm's infields must be won by centuries of clearing stones. The productive land concentrates in a few places: the inland valleys of eastern Norway, the coastal strips and fjord heads of the west, the broad flatlands around the Trondheim Fjord, and above all the Danish plains.

The contrast between the three Scandinavian homelands is stark, and it shaped everything. Denmark is low, fertile, and open — a country of rolling plains where, even today, nearly three-fifths of the land is arable. Sweden sits in the middle: forested and lake-strewn, with its good farmland concentrated in the southern provinces and around Lake Mälaren. Norway is the outlier, a country turned sideways to the

¹Sognefjord dimensions (205 km long, 1,308 m deep; longest and deepest in Norway, second-longest in the world) from the peer-reviewed study of its deep-water benthic communities: Anna Metaxas et al., "Drivers of Megabenthic Community Structure in One of the World's Deepest Silled-Fjords, Sognefjord (Western Norway)," *Frontiers in Marine Science* 7 (2020), art. 393. The fjord's seaward sill lies at 170 m depth, stratifying its water masses.

²"Only about 3% of the land surface is suited for cultivation or arable farming" (Norway): Mons Kvamme, "Mountain Farming of Norway: Land use history and development of cultural landscapes," Food and Agriculture Organization of the United Nations, 2009.

sea, where farming is a marginal activity carried on in defiance of the terrain. The table below gives the modern figures — not a Viking Age census, but the geology has not changed, and the proportions tell the story:³⁴

Region	Arable land (% of land area)	Character of the farmland
Denmark	59.1	Low rolling plains; the breadbasket of the Norse world
Sweden	6.2	Southern plains (Skåne), Mälaren valley; vast forest interior
Norway	2.2	Inland valleys, fjord heads, coastal strips — fragments between rock and water
Iceland	1.2	Thin coastal lowlands; no forests to clear, thin volcanic soils

*Table 1. Arable land as a share of total land area, 2023 (World Bank data); Norway’s long-run figure of ~3% suited to cultivation (FAO). Modern numbers, ancient geology.*⁵⁶

A farm in Jutland and a farm in a western Norwegian fjord operated within very different ecological limits. Jutland offered broad arable fields; many western Norwegian farms occupied narrow strips of workable land between rock and water. Where cultivable acreage could not easily expand, fishing, boating, exchange, seasonal movement, and maritime access carried greater economic weight.

³Arable land as % of land area, 2023, World Bank development indicators: Denmark 59.1, Sweden 6.2, Norway 2.2, Iceland 1.2. Via TheGlobalEconomy.com rankings. Modern figures; the underlying geology is unchanged since the Viking Age, though forest clearance, drainage, and modern methods have altered the farmed area.

⁴On the regional character of Norse farmland — inland valleys, coastal strips, fjord heads, Danish plains — see the standard syntheses: Else Roesdahl, *The Vikings* (Penguin, rev. ed.); Peter Sawyer, *The Viking Age* (in *Kings and Vikings*, Methuen).

⁵Arable land as % of land area, 2023, World Bank development indicators: Denmark 59.1, Sweden 6.2, Norway 2.2, Iceland 1.2. Via TheGlobalEconomy.com rankings. Modern figures; the underlying geology is unchanged since the Viking Age, though forest clearance, drainage, and modern methods have altered the farmed area.

⁶“Only about 3% of the land surface is suited for cultivation or arable farming” (Norway): Mons Kvamme, “Mountain Farming of Norway: Land use history and development of cultural landscapes,” Food and Agriculture Organization of the United Nations, 2009.

Where the Farms Could Be

The Norse farm was not placed on the land so much as negotiated with it. In western Norway the good sites were the fjord heads, where rivers had spent millennia laying down fans of silt, and the occasional flat terrace where an old shoreline had been lifted by the rebounding land. A farm needed, in rough order: a patch of infield flat enough to till, meadow for winter hay, outfield grazing, access to the forest for timber and firewood, and — in the west — a foreshore where a boat could be hauled up. Miss any one of these and the farm failed. The result was a settlement pattern of scattered single farmsteads, each a small, complete world, strung like beads along the water rather than clustered into villages. Neighbours might be an hour's row apart. The fjord was the street.

Eastern Norway offered more room: the long inland valleys — Gudbrandsdalen, Valdres, Hallingdal — with their valley-bottom farms and vast upland commons, and the flatter districts around the Oslofjord. Here farms could be larger, the forest economy richer, and the distance to the sea greater; the inland valleys bred the forest farmers of the inland world we'll meet below. Sweden's geography added a third pattern: the great lake districts of the interior, with farms strung along lake shores and river valleys, and the fertile southern plains of Skåne, which geologically and agriculturally belong more to Denmark than to the Norse north. Sweden's long Baltic shore faced east, toward a different sea and a different set of opportunities — more on that below.

Denmark, finally, was the country the ice had treated kindly. Low, rolling, and ice-smoothed rather than ice-gouged, it offered the only truly extensive farmland in the Norse world — the reason Denmark could support a denser population, larger political units, and the kind of surplus that builds kingdoms. The Danish islands, the Jutland peninsula, and Skåne formed a single agricultural zone, and its farmers lived closer together, in a more village-like pattern, than their Norwegian cousins. When Danish kings mustered the great fleets of the later Viking Age, it was this surplus — grain, men, horses — that paid for them.

A Climate Without Simple Causes

Every popular account of the Vikings used to contain the same neat story. You know the one: the climate warmed, the ice retreated, the Norse sailed west to a green and welcoming Greenland, and then the cold returned and the settlements withered. It is a satisfying story. It is also — and this is the mystery worth following — too simple to be true. The way it fell apart is a lesson in how Viking Age history is actu-

ally written: not from sagas alone, but from tree rings, lake mud, and glacial moraines.

The suspect had a name: the Medieval Warm Period, or Medieval Climate Anomaly, conventionally dated to roughly 950–1250 CE — squarely overlapping the Viking Age. For decades it was treated as a global phenomenon, a medieval analogue of modern warming that explained Norse expansion the way a warm spring explains a good harvest. That picture has not survived. The current understanding, built from dozens of regional proxy records, is that medieval warmth was patchy: some regions were warmer, others were not, and the timing varied from place to place.⁷

Exhibit one: Greenland, the centrepiece of the old story. In the standard telling, Erik the Red named it “Greenland” because it was green — a marketing trick for a warm land. But Nicolás Young’s team at Columbia’s Lamont-Doherty Earth Observatory dated glacial moraines in southwestern Greenland and found something startling: between 975 and 1275 — precisely the centuries of the Norse settlement — the glaciers stood near their Little Ice Age maximum positions. In other words, when the Norse arrived in Greenland, it was at least as cold as it was when they left, centuries later, in the depths of the Little Ice Age. The “warm Greenland” of the textbooks was, at least in that region, a mirage.⁸

Exhibit two: the tree rings. Jesper Björklund and colleagues at the Swiss Federal Institute for Forest, Snow and Landscape Research analysed 188 Scots pines from Sweden and Finland — some fifty million individual tree cells — to reconstruct summer temperatures across Scandinavia for the past 1,200 years. Their finding, published in *Nature*, was blunt: Scandinavia is warmer now than at any point in the past twelve centuries, and the medieval warm phase “was not as warm as previously thought.” The Vikings did not live through anything resembling today’s warmth; they lived through something much closer to the long-term average, with cold snaps and warm spells layered on top.⁹

But a good detective doesn’t stop at two exhibits. And yet — and this is why the evidence does not support either a simple “there was medieval warming” or “there was none”; it depends where you look — there is

⁷The Medieval Climate Anomaly, conventionally c. 950–1250, is now understood as regionally patchy rather than globally synchronous; see the synthesis in the IPCC AR6 Working Group I report on last-millennium climate, and the discussion in Nicolás Young’s Greenland work below.

⁸Nicolás E. Young et al., “Deglaciation of coastal south-western Greenland,” *Science Advances* (2021): glacial moraines show ice near Little Ice Age maximum positions between 975 and 1275 CE.

⁹Jesper Björklund et al., “High-resolution temperature reconstruction of the past 1,200 years in Scandinavia,” *Nature* (2023): 188 Scots pines from Sweden and Finland, ~50 million tree cells analysed; Scandinavia is warmer today than at any point in the past 1,200 years; the medieval warm phase “was not as warm as previously thought.”

genuine counter-evidence. Yarrow Axford's team at Northwestern University cored lake sediments in the Norse Eastern Settlement of Greenland and read the oxygen isotopes in chironomid (lake-fly) remains, a sensitive thermometer for summer temperature. They found a brief warm interval interrupting the long cooling trend: near its end, summers were about 1.5°C warmer than in the surrounding cooling centuries — comparable to southern Greenland summers today. But Axford was careful about the scope: "Unlike warming over the past century, which is global, Medieval warmth was localized." A warm pocket in one fjord district of Greenland is not a global climate epoch.¹⁰

So what does a history of the Viking Age do with this? The responsible position is the one the evidence forces: scholars disagree, regional proxies differ, and no single climate curve explains the Norse expansion. The old monocause — warm weather lured the Vikings outward — is no longer tenable. What replaced it is more interesting. The Norse expansion happened *despite* a climate that was, in Greenland at least, no warmer than the cold centuries that followed. That means the causes must be sought elsewhere: in ship technology, in population pressure on scarce farmland, in politics, in trade, in the simple human fact that the sea was there. Climate mattered — it always matters to farmers living at the margin — but as one instrument in the orchestra, not the whole score. When this book discusses the Greenland settlements' end, it will not be telling a story about weather. It will be telling a story about choices made in a climate that was never as kind as the legend claimed.

Two Viking Worlds: Coast and Inland

To speak of "the Vikings" as one people in one landscape is a convenience that hides a real division. Within Norway and Sweden there were, broadly, two Viking worlds, and they faced in opposite directions. To see them clearly, meet someone. Call him Einar.

Einar is a composite, a name given to the countless boys raised on farmsteads along a thousand miles of shore. He is twelve, wiry from rowing, and he knows the water the way other children know fields. This morning, before the sun, he is in the stern of his father's boat on the inner lead, watching the landmarks slide by the way other boys watch fences pass a cart. He can read the colour that means shallows, the ripple that means a reef, the cloud over that headland that means weather

¹⁰G. Everett Lasher, Yarrow Axford et al., chironomid oxygen-isotope study of the Norse Eastern Settlement, Greenland: a brief warm interval interrupted the long cooling trend, reaching ~1.5°C above surrounding centuries near its end — comparable to modern southern Greenland summers (~10°C). Axford: "Unlike warming over the past century, which is global, Medieval warmth was localized."

coming. He is not a warrior. He is not a trader. He is something the sagas rarely name but always assume: a coastal farmer's son, for whom the sea was never an adventure. It was the road.

His world is the first Viking world. The coastal farmer lived with his back to the land and his face to the water. His farm sat where the fjord or the skerry shore gave him a patch of infield, but his true field was the sea: the fishing grounds, the seal skerries, the tidal currents where the cod ran. He kept a boat the way an inland farmer kept a plough — it was not a luxury but the basic tool of the economy. Fish, dried and traded, could buy what the thin soil could not grow. News, marriage partners, and political power all arrived by water. The coastal world was open, mobile, and outward-looking; its horizons were measured in sailing days, not walking hours. When the sagas describe chieftains gathering fleets or merchants sailing to Hedeby and beyond, they are describing the logic of this world extended to its fullest.

Now turn inland, and meet Einar's opposite. The inland farmer lived a different life, and it is the one most easily forgotten because it left fewer ships and fewer sagas. In the valleys of eastern Norway and the forest districts of Sweden, the farm faced inward: toward the infield, the outfield grazing, and above all the forest. The forest was a working landscape — pasture for livestock, source of timber, firewood, charcoal, and iron from bog ore, hunting ground for elk and fur-bearing animals whose pelts were themselves a trade good. The inland economy ran on a seasonal rhythm that the coast shared but did not depend on in the same way: the shieling system. Each summer the livestock were driven up to summer dairy farms — the *sel* or *seter* — at higher elevations, where the milk cows and ewes could be kept close for daily milking and the milk processed on site into butter and cheese. The herds typically went up in mid-June and came home in mid-October, and the whole household's calendar turned around that migration.¹¹ Archaeology has caught this rhythm in the act — a working summer farm in the middle of the Viking Age, exactly where the system says it should be.¹²

The forest itself was an economy, not an emptiness. From its bogs came bog iron, smelted in small furnaces into the iron every farm needed for tools and weapons. From its trees came timber for houses and ships, firewood, and charcoal. From its animals came meat, hides, and above all furs — squirrel, marten, beaver, elk-hide — which were among the most valuable trade goods the inland world produced, worth carrying over long distances to coastal markets and beyond. The forest

¹¹Shieling (*sel/seter*) practice — livestock driven to summer dairy farms at higher elevation, typically mid-June to mid-October, milk processed on site — from standard Norse rural-economy scholarship; cf. Hurstwic's survey of the archaeological and textual evidence.

¹²Søstølen, Eidsdal valley, Møre og Romsdal: Viking Age shieling with charcoal dated AD 882–1015, reported in *Remote Sensing* (MDPI), 2025.

was also a social fact: a frontier where the law ran thin, where outlaws lived, where new farms could be cleared by men with no inheritance — the younger sons for whom the divided infield at home had nothing left. Much of the internal colonization of the Viking Age, the slow push of settlement into the forest margins of Sweden and eastern Norway, was driven by exactly these land-hungry men.

These were not two separate peoples but two poles of a connected economy, and many farms drew on both. A coastal farm still kept cattle and cut hay; an inland farm could send surplus downriver toward coastal exchange. The balance differed by region, shaping the mix of skills, resources, and routes available to each household. The inland farmer's wealth was in land, livestock, and forest rights — things that could be divided among sons until there was nothing left to divide. The coastal farmer's wealth was in boats, nets, and the knowledge of waters — things that could not be subdivided, and that pointed outward. It is no accident that the great outward movements of the Viking Age — the westward voyages, the raiding fleets, the trading ventures — were overwhelmingly a coastal phenomenon. The men who sailed to Iceland and beyond were not, for the most part, the sons of the deep forest. They were the sons of the shore, for whom the sea had always been the road.

Denmark differed from the fjord districts in its greater extent of arable land and denser, more village-oriented settlement. Agriculture therefore carried a different weight in its economy, while royal power and military organization developed within the resources and communication networks of the Danish islands, Jutland, Skåne, the North Sea, and the Baltic.

The Sea as Highway

If the land gave the Norse little, the sea gave them everything — but only because of a particular combination of physical facts that made the Norwegian coast one of the great natural routeways of the medieval world.

The first fact is the coast itself. Norway's shoreline, measured around every fjord, bay, and island, runs to some 25,148 kilometres — enough coastline to stretch from Oslo to New Zealand and most of the way back, folded into one long, narrow country.¹³ Much of that length is not open shore but a maze: the *skjærgård*, the “skerry garden,” a chain of off-shore islands, islets, and reefs running along most of the coast. Between

¹³Norway's coastline: 25,148 km (mainland, fjords, bays, and islands at standard cartographic resolution), FAO Fishery Country Profile: The Kingdom of Norway (FID/CP/NOR, 2005). Higher-resolution surveys of the mainland-plus-islands coastline exceed 100,000 km — the classic coastline paradox; figures vary with measurement scale.

the skerries and the mainland lie the inner leads — sheltered channels where a ship can sail for hundreds of kilometres in calm water while the Atlantic breaks itself on the outer rocks. This was the Norse highway system. Coastal voyaging meant threading the leads from landmark to landmark, beaching at night, never losing sight of land unless you chose to. It was how fish, timber, news, and armies moved. A chieftain in Trøndelag could reach the markets of the south without ever putting his ship's bow to the open ocean.

The second fact is the warm water. The North Atlantic Current — the great northeastward extension of the Gulf Stream — carries tropical warmth up the Norwegian coast, and its effect is one of the most consequential climatic facts for coastal Norway. Lofoten, the archipelago off northern Norway, sits at 68° north — the latitude of central Greenland, of the northernmost mainland of Alaska. By rights it should be locked in Arctic ice half the year. Instead, the current keeps its winter temperatures around freezing, near 0°C, and its harbours workable through the dark months.¹⁴

Place	Latitude	Winter character	Why it differs
Lofoten, Norway	68°N	Around 0°C; harbours open	North Atlantic Current
Central Greenland	68°N	Deep Arctic cold; continental ice	No warming current
Northern Alaska coast	~68–70°N	Arctic winter; sea ice	Polar currents, continental cold

*Table 2. The latitude paradox: three places at roughly the same latitude, three different winters. The current is the difference between a working coast and a frozen one.*¹⁵

The third fact is what the cold-and-wind did to fish. Lofoten's climate — cool summers, near-constant wind, winter temperatures hovering around freezing — turned out to be ideal for a particular kind of alchemy: air-drying cod into stockfish, without salt and without smoke. The fish were gutted, hung on wooden racks, and left to the wind for months; what came down in spring was a hard, light, nearly indestructible protein that would keep for years and could be shipped anywhere.¹⁶ This was the economic engine that turned geography into

¹⁴Lofoten at 68°N with winter temperatures around 0°C, warmed by the North Atlantic Current; the current-plus-wind regime as the basis of stockfish production: Lofotr Viking Museum / Museum Nord material on the Lofoten fisheries.

¹⁵Lofoten at 68°N with winter temperatures around 0°C, warmed by the North Atlantic Current; the current-plus-wind regime as the basis of stockfish production: Lofotr Viking Museum / Museum Nord material on the Lofoten fisheries.

¹⁶Stockfish: cod air-dried on wooden racks without salt or smoke, exploiting Lofoten's cool summers and near-constant wind; the dried product keeps for years and became a major medieval trade commodity. Lofotr Viking Museum / Museum Nord.

wealth. The Norwegian coast could not grow grain in quantity, but it could harvest the sea in quantities no farmland could match, preserve the harvest with nothing but air and patience, and trade it for everything the land lacked. Stockfish would become one of the great commodities of medieval Europe — the dried cod of Lofoten feeding cities hundreds of miles away, the profits flowing back up the leads to the farms and chieftains of the coast.

Stockfish deserves a closer look as a form of wealth, because it shaped the coastal economy in ways grain never could. Grain rots; stockfish endures. A chieftain could stockpile dried cod the way a king stockpiles silver — it was storable, divisible, transportable value, and it could be converted into ships, into weapons, into the loyalty of followers. It could also be eaten on the voyage itself: a ship provisioned with stockfish carried its food supply in a form that would not spoil crossing the Atlantic, a logistical fact that underwrote the westward voyages as surely as any sail. The Norse did not invent dried fish — coastal peoples everywhere preserve their catch — but the Lofoten climate let them produce it at a scale and quality that became an export industry, and the inner leads let them move it to market. Geography, technology, and trade interlocked: the current made the fish dryable, the skerries made them shippable, and the hunger of distant markets made them valuable.¹⁷

The sea was not just a road. It was the farm that never failed.

The Eastern Sea

Everything said so far describes Norway's Atlantic coast. Sweden faced a different sea, and it made a different kind of Viking.

The Baltic is the Atlantic's opposite: shallow, brackish, nearly enclosed, strewn with islands. Where the Norwegian coast offers the inner leads, the Baltic offers stepping stones — the Åland islands, Gotland, Öland — breaking long crossings into short, manageable stages across sheltered water. A Swedish farmer-trader sailing east island-hopped in the same way his Norwegian cousin lead-hopped, and arrived at the mouths of the great river systems that drain the eastern continent. From there the road ran south by river, not by sea: the route the Norse called the *Austrvegr*, the east-way, down toward the markets of the Rus lands, the Caspian, and Byzantium. It was on the Volga that the Arab traveller Ibn Fadlan met the Rus in 921–22 and left his famous description of them — tall as date palms, in Richard Frye's rendering — which

¹⁷Stockfish: cod air-dried on wooden racks without salt or smoke, exploiting Lofoten's cool summers and near-constant wind; the dried product keeps for years and became a major medieval trade commodity. Lofotr Viking Museum / Museum Nord.

reminds us that the eastern road was as much a part of the Viking world as the western one.¹⁸

The Baltic shaped its sailors accordingly. Its waters freeze in hard winters, its storms are short and vicious rather than long and oceanic, and its trade ran on bulk goods — furs, wax, honey, iron, slaves — moving between the forest zone and the wealthy south. The Swedish Viking Age was a riverine and mercantile age as much as a maritime one, and its geography explains why: Sweden's coast looks east, its rivers run east, and the sea beyond them is a lake the size of a country.

Geography did not make the Norse fearless sailors by nature. It made them practised ones. Think of Einar again — the boy in the stern, reading the water. A boy raised on the inner lead learned to read water the way other children learned to read fields: the colour that means shallows, the ripple that means a reef, the cloud over that headland that means weather coming. By the time such a man put his ship's bow toward Iceland, he had a lifetime of seamanship behind him. The open Atlantic was a different proposition — but it was a proposition made by men who had spent their lives on the water.

The Open Atlantic and Its Perils

The inner leads made the Norse masters of the coast. The open Atlantic made them gamblers.

West of the skerries, the protections fall away one by one. There are no landmarks — only the sun, the stars, the flight of birds, and the colour of the water. There is no beach for the night; a ship caught by a North Atlantic gale rides it out or it does not. The crossings the Norse undertook — to the Faroes, to Iceland, to Greenland, and on one famous occasion to the coast of North America — were voyages across hundreds of miles of open water in small, open-decked ships with low freeboard and little shelter. They succeeded often enough to change history, which tells us how good the sailors were; the sagas' wreck stories, and the ships that simply never arrived, tell us the price.

The westward route formed a chain of stepping-stone destinations — Shetland, the Faroes, Iceland, then Greenland — but the major legs were open-water crossings, not islands continuously visible from one another. Shetland lies roughly 280 kilometres south-east of the Faroes, and the Faroes sit midway between Scotland and Iceland: distances that

¹⁸Ibn Fadlan, *Risala* (921–22), on the Rus at the Volga: “I have never seen more perfect physical specimens, tall as date palms, blond and ruddy” — Richard N. Frye's translation, *Ibn Fadlan's Journey to Russia* (Markus Wiener, 2005). The eastern river routes (*Austrvegr*) are the standard scholarly framework for Swedish Viking Age expansion eastward; cf. the syntheses of Sawyer and Roesdahl.

demanded genuine blue-water navigation.¹⁹ Each island was a place to reprovision, to wait out weather, to repair. But each was also a point of no return. Past Iceland there is no rescue, no friendly shore — only the destination or the sea. And Greenland itself, the farthest stone, sat at the mercy of the drift ice that came down its east coast from the Arctic. In cold years the ice closed the harbours and cut the settlements off from Iceland and Norway for seasons at a time; the lifeline that carried timber, iron, and news was a thread, and the ice could snap it. The Norse knew this and went anyway — which is either courage or desperation, and was probably both.

The dangers were not only storms. Fog could swallow a fleet in an hour. Drift ice came down from the Arctic along the Greenland coast — the very ice that made the Norse Greenland settlements' position so precarious, locking their harbours and cutting their lifeline to Iceland and Norway. Currents could carry a ship far off its reckoning. Norse navigation relied on accumulated seamanship: sun and stars when visible, winds and swell, birds, water colour and depth, remembered sailing directions, and the reading of coast and cloud. Polarising crystals proposed as navigational “sunstones” are physically plausible and experimentally intriguing, but their use aboard Viking Age ships remains a hypothesis rather than a securely attested technique.²⁰ What the geography gave with one hand, the ocean took with the other. The sea was a highway along the coast and a wilderness beyond it, and the Viking Age was built by men and women who learned to travel both.

Geography as First Cause

Stand again on that fjord at dawn. The ship slips between the skerries. The walls rise on either side. The little green tongue of farmland waits at the water's head.

Everything in this book flows from that scene.

The fjords and fjells made farmland scarce, and scarce farmland made the Norse look outward. The forests gave them timber for ships and charcoal for iron; the shielings taught them to move with the seasons. The climate — neither the warm Eden of the old textbooks nor the

¹⁹Scottish Government, *Shetland Islands Regional Marine Plan* (2025), gives Shetland as approximately 280 km south-east of the Faroe Islands. The Government of the Faroe Islands describes the archipelago as lying midway between Scotland and Iceland. These distances rule out continuous visual island-hopping on the major North Atlantic legs.

²⁰The “sunstone” navigation hypothesis is experimentally plausible but archaeologically unproven for the Viking Age. See Philip Ball, “A light compass?” *Nature Materials* 10 (2011), 909, and the optical work it discusses. A calcite crystal associated with navigation was recovered from a much later, sixteenth-century wreck; it does not demonstrate Viking Age shipboard use. Chapter 13 keeps the same caution.

frozen waste of the legend — was a marginal farmer's climate, workable but unforgiving, and it rewarded exactly the diversified, sea-oriented economy the Norse built. The skerry chains gave them a sheltered highway along the whole length of their world; the warm current gave them open harbours at Arctic latitudes and the wind-dried cod that became their gold. And the open ocean beyond gave them somewhere to go: Iceland, Greenland, the river roads of the east, the coasts of the west.

Geography did not determine the Viking Age. Peoples in hard landscapes do not always take to the sea, and the Norse expansion had political, demographic, and technological causes that no map can show. But geography drew the map on which everything else was played. It made the ship the natural tool, the coast the natural road, and the farm — the subject of our next chapter — a place defined as much by water as by land.

A marginal farm in a short growing season left little room for mistimed work: hay had to be cut within the workable weeks, livestock moved with the grazing cycle, and boats maintained before autumn weather. The annual round joined spring sowing, summer fishing and haymaking, shieling movement, autumn slaughter and provisioning, and winter craft and repair. These seasonal constraints made planning, stored surplus, local knowledge, and timely movement practical necessities for households living across much of the Norse world.

Notes
